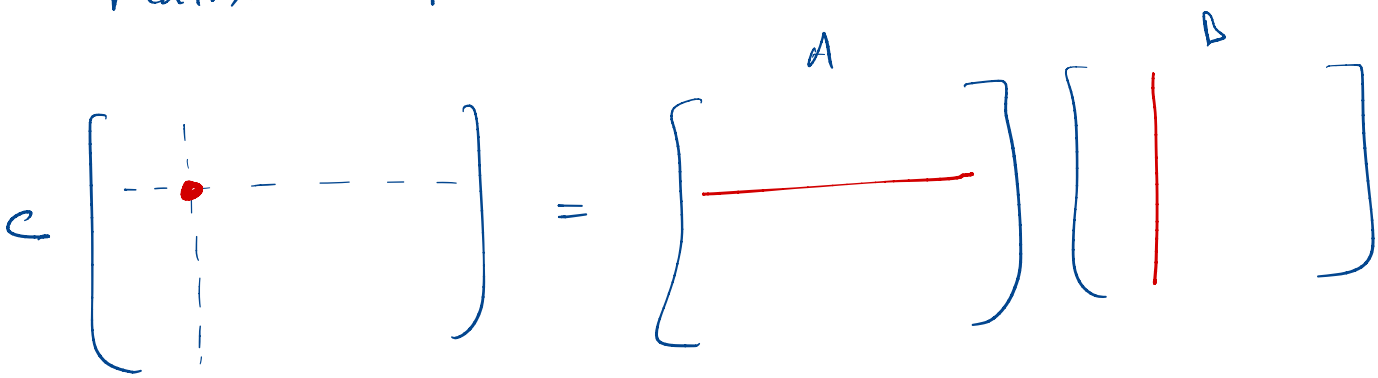
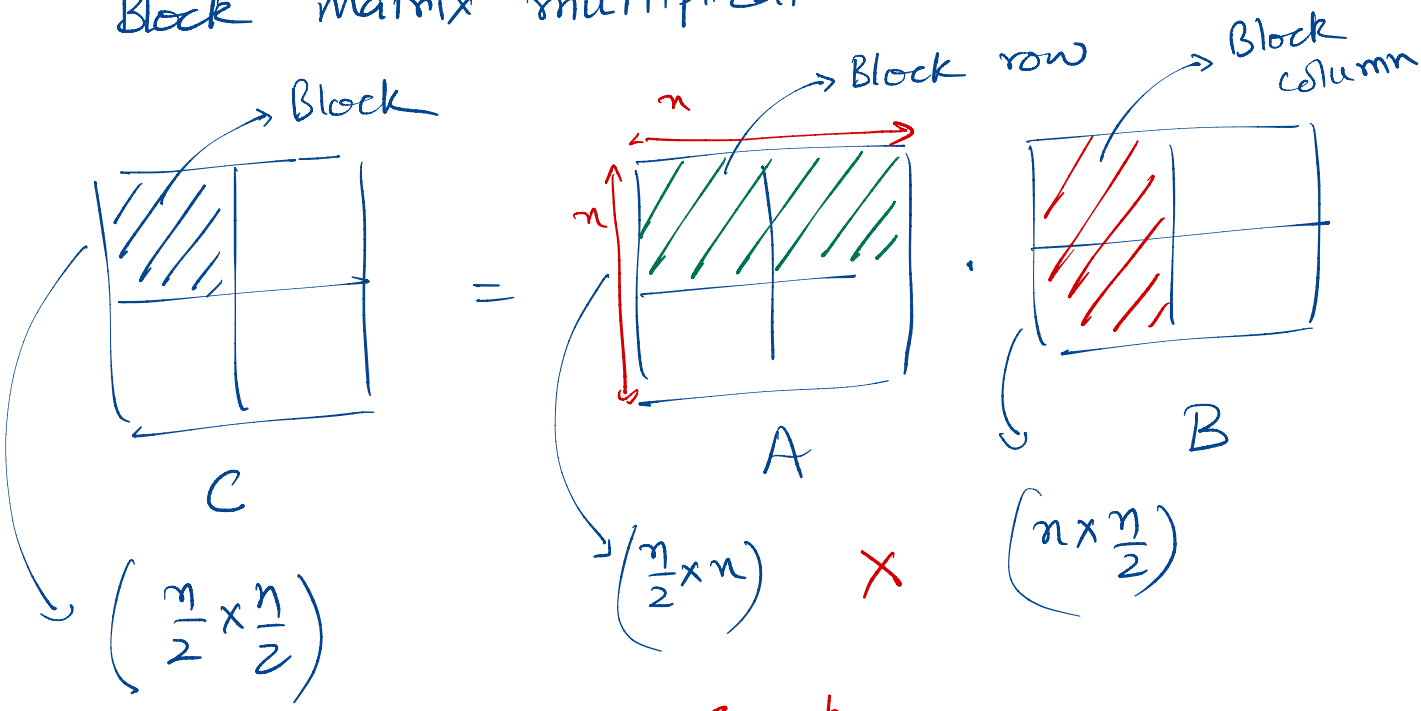
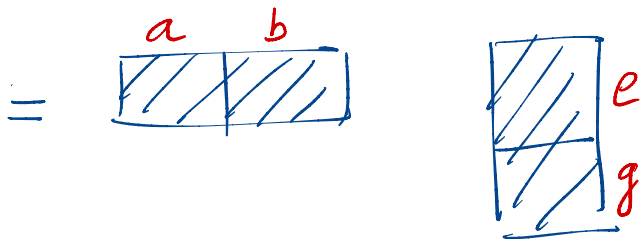
Matrix Multiplication

Matrix multiplication seen as dot products:



Block matrix multiplication





$$= a \cdot e + b \cdot g$$

$$T(n) = 8\,T\left(\frac{n}{2}\right) + \Theta(n^2)$$

$n^{\log_b a}$ $\qquad a = 8$ $\qquad \log_b a = \log_2 8 = 3$

$b = 2$

$\hookrightarrow \underline{n^3}$

$f(n) = \underline{n^2}$

Case I applies

$$\Rightarrow \quad T(n) = \Theta(n^3)$$